Priority	Category	Metric	Simple-English meaning	Value range / low, medium, high / ideal	Formula / measurement	Formula terminology
1	Safety & trust	Sensitive data leakage rate	Does it reveal private information?	0–100%; ideal 0%; any severe case requires review.	Disclosures of protected data / Privacy test cases × 100	Protected data defined by permission and data policy.
2	Safety & trust	Unsafe response rate	How often the AI gives unsafe replies.	0–100%; lower ideal ~0%; inspect severe failures separately.	Policy-violating responses / Safety test responses × 100	Use scenario-labeled safety policy tests.
3	Safety & trust	Consent / disclosure compliance	Does it tell people necessary information?	0–100%; ideal 100% for applicable requirements.	Correct disclosures / Disclosure-required interactions × 100	Disclosures include AI identity or recording consent where applicable.
4	Safety & trust	Escalation / handoff success	Does it transfer to a person when it should?	0–100%; higher better; target near 100% for critical scenarios.	Successful human handoffs / Required handoff scenarios × 100	Required scenarios predefined by policy and task.
5	Experience	End-to-end successful call rate	Did the entire experience work well?	0–100%; higher better; set gates before testing.	Calls meeting task, quality, latency and safety criteria / All calls × 100	A success must pass all predefined gates, not just task answer.
6	Dialogue quality	Task completion rate	Can users actually get things done?	0–100%; higher better; target calibrated by task difficulty.	Successfully completed tasks / Attempted tasks × 100	Success must satisfy scenario-specific objective criteria.
7	Dialogue quality	Hallucination rate	How often it confidently says made-up things.	0–100%; lower ideal ~0%, especially in high-stakes cases.	Unsupported factual claims / All evaluated factual claims × 100	Claims checked against authoritative ground truth.
8	Dialogue quality	Instruction adherence	Does it follow what the user asked?	0–100%; higher better; near 100% for critical constraints.	Compliant responses / Evaluated instruction-bearing turns × 100	Includes format, language, scope and task constraints.
9	Latency & turn-taking	Turnaround time (TAT)	How long the user waits after finishing before AI starts speaking.	≥0 ms; low faster, high slower; provisional ideal p50 <800 ms and p95 <1500 ms for interactive use; benchmark by network and task.	t(first audible AI speech) - t(user speech end)	Time stamps in milliseconds; include endpointing, reasoning, and synthesis.
10	Latency & turn-taking	Time to first audio (TTFA)	How quickly generated audio starts arriving.	≥0 ms; lower better; provisional p50 <500 ms; separate from TAT.	t(first audio packet) - t(request accepted)	Measure at the client; excludes playback buffering unless separately counted.
11	Latency & turn-taking	End-of-turn detection delay	How long AI waits to recognize the user has stopped.	≥0 ms; lower better unless it cuts users off; evaluate jointly with premature endpoint rate.	t(endpoint decision) - t(actual user speech end)	Endpoint decision is when the system commits that the user finished.
12	Latency & turn-taking	Premature endpoint rate	How often AI assumes the person is done while they are still speaking.	0–100%; low ideal ~0%; high bad; report by pause length and speaking style.	Early endpoints / True user turns × 100	Early endpoints incorrectly split an unfinished utterance.
13	Latency & turn-taking	Latency jitter	Whether response speed feels consistent.	≥0 ms; lower better; compare similar scenarios and devices.	SD(TAT) or p95(TAT) - p50(TAT)	SD is standard deviation; percentiles describe variation.
14	Latency & turn-taking	Barge-in stop latency	How quickly AI stops talking when interrupted.	≥0 ms; lower better; provisional <500 ms; test noisy environments.	t(AI audio stops) - t(user interruption onset)	Include detection and playback stopping.
15	Latency & turn-taking	Barge-in detection latency	How fast AI notices someone interrupting it.	≥0 ms; lower better; provisional <300 ms under clean audio.	t(interruption detected) - t(user interruption onset)	Interruption onset is the first audible user speech while AI speaks.
16	Latency & turn-taking	Interruption handling success	Whether users can interrupt naturally.	0–100%; higher better; target ≥95% after validation.	Correctly handled interruptions / All valid interruptions × 100	Correct means stops, listens, and answers the interruption without losing context.
17	Conversational	Turn-taking appropriateness	Does it know when to speak and when to listen?	0–100%; low <70, medium 70–89, high ≥90 illustrative; ideal ≥95% after validation.	Appropriate handoffs / Total handoffs × 100	Appropriate = human-rated timing, including whether AI waits, yields, and resumes correctly.
18	Conversational	Unwanted overlap rate	Does it talk over you unnecessarily?	0–100%; low = good, high = bad; ideal near 0%, excluding acceptable backchannels.	Unwanted overlapping talk time / Total conversation time × 100	Unwanted overlap = both sides speak when interruption/acknowledgment was not appropriate.
19	Latency & turn-taking	Unwanted overlap rate	How often AI talks over the user.	0–100%; lower ideal ~0%; annotate intentional overlap separately.	Unwanted overlapping turns / Total turns × 100	Count AI speech that improperly overlaps a user turn; exclude cooperative backchannels.
20	Speech recognition	Word error rate (WER)	How many words AI mishears.	≥0%, can exceed 100%; lower better; <10% clean speech is a provisional target, compare accents/noise.	(S + D + I) / N × 100	S substitutions, D deletions, I insertions, N reference words.
21	Speech recognition	Entity recognition accuracy	Does AI catch important details correctly?	0–100%; higher better; aim ≥98% for critical fields with confirmation.	Correctly transcribed entities / Total entities × 100	Entities include names, dates, numbers, addresses and IDs.
22	Speech recognition	Character error rate (CER)	How accurately AI hears names and languages without spaces.	≥0%, can exceed 100%; lower better; benchmark per language.	(S_c + D_c + I_c) / N_c × 100	Character-level edit errors and total reference characters.

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23	Speech recognition	Speech detection recall	How often AI notices when someone speaks.	0–100%; higher better; aim ≥98% in target conditions.	Correct user speech detected / All annotated user speech segments × 100	Speech segments annotated by humans.
24	Speech recognition	False speech activation rate	How often background noise wakes the AI.	≥0 per hour; lower better; calibrate to environment.	False speech triggers / Audio hours	Trigger means noise wrongly classified as user speech.
25	Dialogue quality	Intent understanding accuracy	Does it understand what users want?	0–100%; higher better; ≥95% is a pilot target for constrained intents.	Correctly identified intents / Annotated intents × 100	Compare intent inferred by AI to human-labeled user intent.
26	Dialogue quality	Context retention accuracy	Does it remember what was said earlier in the call?	0–100%; higher better; ≥95% for tested context windows.	Correct context-dependent answers / Context probes × 100	Probes require recalling relevant prior turns.
27	Conversational	Contextual response quality	Does it respond like someone following the conversation?	1–5; low 1–2, medium 3, high 4–5; ideal ≥4.	Mean rubric rating (1–5)	Evaluators rate relevance, coherence, continuity, and conversational fit against the full dialogue.
28	Dialogue quality	Clarification appropriateness	Does it ask sensible follow-up questions?	0–100%; higher better; evaluate ambiguity balance.	Correct clarification decisions / Ambiguous or clear test cases × 100	Correctly asks when needed and avoids asking when unnecessary.
29	Dialogue quality	Repair success rate	Can it recover gracefully when it gets something wrong?	0–100%; higher better; target ≥90% pilot.	Successful recoveries / Misunderstanding or error events × 100	Recovery fixes the issue within a defined number of turns.
30	Dialogue quality	Turns to resolution	How many exchanges it takes to solve a problem.	≥1; fewer can be better but not at expense of accuracy or empathy.	Total user + AI turns in completed sessions / Completed sessions	Turn is one conversational contribution.
31	Dialogue quality	Repetition / redundancy rate	Does AI keep repeating itself?	0–100%; lower ideal.	Unnecessary repeated AI turns / AI turns × 100	Human labels redundant content.
32	Dialogue quality	Conversation abandonment rate	How often people give up before finishing.	0–100%; lower better; interpret with task difficulty.	User-abandoned sessions / Started sessions × 100	Exclude intentional quick exits; segment by reason.
33	Acoustic / quality	Pronunciation accuracy	Does it pronounce names and words correctly?	0–1; low <0.9, medium 0.9–0.97, high >0.97 illustrative; ideal near 1.	1 − (Pronunciation errors / Evaluated words)	Error = independently annotated incorrect pronunciation; evaluated words = all assessed words.
34	Speech output	Synthesis artifact rate	How often generated speech sounds broken.	0–100%; lower ideal ~0%.	Clips with audible glitches / All generated clips × 100	Glitches include clicks, pops, dropouts, robotic breaks.
35	Speech output	Audio dropout rate	How often the voice cuts out.	≥0 per minute; ideal ~0.	Missing or corrupted playback intervals / Playback minutes	Count missing chunks or audible gaps from transport/synthesis.
36	Speech output	Clipping rate	Does the voice distort at loud moments?	0–100%; ideal ~0%.	Clipped audio samples / Total samples × 100	Clipped samples hit digital amplitude limits.
37	Speech output	Voice consistency	Does it keep sounding like the same speaker?	1–5; low 1–2, medium 3, high 4–5; ideal ≥4.	Mean listener consistency rating (1–5)	Same voice identity, accent and timbre across turns; use blind ratings.
38	Speech output	Loudness consistency	Does volume jump unexpectedly?	≥0 LU; lower variation generally better; avoid flattening intentional expression.	SD(integrated loudness per turn, LUFS)	LUFS is loudness relative to full scale; compare turns in same setup.
39	Speech output	Speaking-rate deviation	Does it speak too fast or too slowly?	≥0 syllables/sec; closer to matched human range ideal; no universal cutoff.	abs(AI syllables/sec - reference syllables/sec)	Reference matched by language, content and context.
40	Perceptual	Prosody naturalness MOS	Overall, does the *way* it speaks sound human?	1–5; low 1–2, medium 3, high 4–5; ideal ≥4, with confidence intervals.	Σ listener prosody ratings / Number of ratings	MOS = mean opinion score; blind listeners rate pitch, stress, rhythm, pauses together on a 1–5 scale.
41	Acoustic / prosody	Intonation contour similarity	Does pitch rise and fall at sensible points in the sentence?	≥0; low cost = closer to reference, high = more different; ideal low, but valid alternatives may differ.	DTW(z(F0_AI), z(F0_human))/L	DTW = dynamic time warping alignment cost; z = normalized pitch contour; L = aligned path length. Use several valid human references.
42	Acoustic / prosody	Emphasis / stress accuracy	Does the voice emphasize the important word?	0–100%; low <60, medium 60–84, high ≥85 illustrative pilot bands; ideal ≥90% after validation.	Correct prominent targets / Annotated targets × 100	Annotators mark which words should receive contrastive or informational prominence; correct = listener-verified placement.
43	Acoustic / prosody	Expressive appropriateness	Does it sound cheerful, calm, serious, or empathetic at the right time?	1–5; low 1–2, medium 3, high 4–5; ideal ≥4.	Mean listener rating (1–5)	Listeners rate whether emotional delivery matches intended context; average across raters.

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44	Acoustic / prosody	Pause placement F1	Does it pause where a person would naturally pause?	0–1; low <0.6, medium 0.6–0.84, high ≥0.85 illustrative; ideal close to 1.	2PR/(P+R); P=TP/(TP+FP); R=TP/(TP+FN)	TP = appropriate pauses detected; FP = inappropriate pauses; FN = expected pauses missed.
45	Acoustic / prosody	Articulation rate	How fast the voice speaks while actually talking.	>0 syllables/s; low = slow, high = fast; ideal match matched human and listener comprehension.	Spoken syllables / speaking time excluding silence	Speaking time excludes annotated silent pauses; syllables from aligned transcript.
46	Acoustic / prosody	Pause duration	Are pauses comfortably long, not awkwardly short or long?	≥0 s; low = short, high = long; ideal depends on clause boundary and task.	Σ pause durations / number of pauses	A pause is silence exceeding a fixed threshold (e.g., 200 ms); same threshold across systems.
47	Robustness	Noise robustness delta	How much background noise hurts understanding.	Can be negative; near 0 ideal; smaller positive gap better.	WER(noisy) - WER(clean)	Matched utterances with and without noise; percentage-point gap.
48	Robustness	Accent performance gap	Does it work equally well across accents?	≥0 percentage points; lower ideal; report each group separately.	max(group error rate) - min(group error rate)	Compare adequately sampled accent groups on matched tasks.
49	Robustness	Multilingual / code-switch accuracy	Can it handle users switching languages?	0–100%; higher better; validate each language pair.	Correct multilingual turns / Tested multilingual turns × 100	Include switches within sentences and across turns.
50	Robustness	Session failure rate	How often calls break entirely.	0–100%; ideal ~0%.	Crashed or unrecoverable sessions / All sessions × 100	Includes disconnected or frozen sessions.
51	Robustness	Recovery after network interruption	Does it recover from connection problems?	0–100%; higher better; target ≥95% for supported reconnects.	Successfully resumed sessions / Interrupted sessions × 100	Resume means restored usable conversation after connectivity returns.
52	Perceptual	Overall voice naturalness MOS	Does the voice feel natural overall?	1–5; low 1–2, medium 3, high 4–5; ideal ≥4.	Σ overall naturalness ratings / Number of ratings	Blind listeners score complete clips/interactions; keep the scale and listening setup consistent.
53	Perceptual	A/B preference win rate	Which of two voices do people actually prefer?	0–100%; <50 loses, 50 parity, >50 wins; ideal statistically reliable >50% against baseline.	AI-A preferred / Non-tied A/B judgments × 100	Blind paired comparisons of identical prompts; report ties separately.
54	Perceptual	Human-likeness rating	How much does it sound like a person rather than a machine?	1–5; low 1–2, medium 3, high 4–5; ideal ≥4, where human-likeness is the product goal.	Σ human-likeness ratings / Number of ratings	Blind rating 1–5 specifically for perceived human-likeness, distinct from usefulness or correctness.
55	Experience	User satisfaction (CSAT)	Did users like the interaction?	0–100%; higher better; target benchmarked to comparable service.	Positive survey responses / All valid survey responses × 100	Positive typically top-two boxes on a five-point scale.
56	Operations	Availability / uptime	Is the voice assistant reliably available?	0–100%; higher better; SLA-dependent, e.g. ≥99.9%.	Available service minutes / Scheduled service minutes × 100	Service must accept and complete requests under health criteria.
57	Operations	Cost per completed conversation	How much does a successful call cost?	≥0 currency; lower better at fixed quality; no universal ideal.	Total voice system cost / Successfully completed conversations	Include ASR, LLM, TTS, infra and telephony where relevant.